

Block	What it does	Input -> output	Responsible file(s)	Side/status
Camera/video	Decodes every frame of each listed training clip.	Video -> RGB frame	76_extract_fs1105_features.py	Python, IMPLEMENTED
Perception	Runs MediaPipe Holistic.	RGB -> 33 pose + 21 right-hand landmarks when detected	76_extract_fs1105_features.py	Python, IMPLEMENTED

Block	What it does	Input -> output	Responsible file(s)	Side/status
Feature builder	Anchors values at the pose nose, optionally zeros missing hand, scales the hand, damps Z.	landmarks -> (162,) float32	build_frame_features()	Python, IMPLEMENTED
Temporalizer	Makes overlapping fixed-length examples.	frame vectors -> (20,162) windows	process_video()	Python, IMPLEMENTED
Dataset/audit	Persists arrays/metadata and validates shapes, finite values, and ownership.	.npy/.meta.json -> reports	76_extract_fsl105_features.py, 75_audit_fsl_dataset.py	Python, IMPLEMENTED
Temporal classifier	Intended residual dilated TCN.	[B,20,162] -> [B,C] softmax	78_train_fsl_rdtcn.py	Python, EXPERIMENTAL; NOT TRAINED
Recognition/output	No FSL live runtime/model asset is wired.	probabilities -> accepted FSL token	none	NOT IMPLEMENTED

Current Android runtime architecture (implemented code, blocked deployment)



Block	Input -> output	Important contract	File(s)	Status
Camera/landmarks	ImageProxy -> LandmarkFrame	front camera mirror policy; pose/hand task assets	VoxGestCameraRecognitionController.kt, LandmarkExtractor.kt	IMPLEMENTED
Feature building	landmarks -> (162,)	99 pose + 63 selected hand	OneHand162FeatureBuilder.kt	IMPLEMENTED
Temporal buffer	vectors -> (20,162)	drops oldest frame when full	LandmarkSequenceBuffer.kt	IMPLEMENTED
TFLite recognition	[1,20,162] requested -> probabilities	labels WHAT, YOUR, NAME, MY, NOTHING	VoxGestTfliteRecognizer.kt	IMPLEMENTED code; BLOCKED by bundled 30-frame model
Acceptance	raw result -> accepted/rejected token	confidence, margin, hand presence, stable consecutive windows, duplicate/cooldown rules	DynamicWordAcceptanceGate.kt	IMPLEMENTED for five-label onehand profile
Presentation	accepted token -> UI/history/TTS/avatar path	raw/rejected predictions must not alter sentence output	controller and Android UI/avatar classes	IMPLEMENTED/partly UNVERIFIED live

The FSL branch is **not** the Android model currently loaded by this source. Current Android profile selection only allows `onehand162_phrase_v1` or `onehand162_android_calibrated_v1`; both use five labels and manifest a 20-frame request.

3. Feature engineering

Implemented FSL-105 feature vector

`76_extract_fsl105_features.py:build_frame_features()` implements:

- Pose: 33 landmarks x (x,y,z) = **99** values.
- Hand: **right hand** only, 21 landmarks x (x,y,z) = **63** values. A missing right hand becomes 63 zeros; a missing pose zeros the complete 162-vector because no safe nose reference exists.
- Anchor: pose and hand coordinates have pose landmark 0 (nose) subtracted. The first pose triplet is then explicitly zero.
- Scale: after nose anchoring, every selected hand coordinate is divided by the Euclidean distance between hand wrist (0) and middle-finger MCP (9), only when scale > 0.001. Pose is not divided by this hand scale.
- Depth: every Z component (2::3) is multiplied by **0.3**.
- Concatenation: `[pose_values.flatten(), hand_values.flatten(), float32, shape (162,)]`.

The Android `OneHand162FeatureBuilder.kt` implements the same high-level recipe, including selected-hand mapping, pose masking, wrist-to-MCP scaling, first-triplet zeroing, and Z damping.

Implemented historical/Android policy distinction

`lstm_features.py` also builds OneHand162, but its extracted dynamic frames are nose-relative and support single-hand pose masking. It does **not** apply the wrist-to-MCP scale or Z damping visible in the FSL extractor/Android builder. The controller's diagnostic text still says `normalization_policy=PYTHON_NOSE_RELATIVE_NO_WRIST_SCALE`; this conflicts with `OneHand162FeatureBuilder.kt` and must not be treated as current parity evidence.

Item	Status	Evidence/meaning
99 pose + 63 hand = 162	IMPLEMENTED	FSL extractor and Android builder
Nose-relative translation	IMPLEMENTED	FSL extractor and Android builder
Wrist-to-MCP hand scale	IMPLEMENTED in FSL/Android	<code>76_extract_fs1105_features.py</code> , <code>OneHand162FeatureBuilder.kt</code>
Z damping x 0.3	IMPLEMENTED in FSL/Android	same files
Single-hand pose masking	IMPLEMENTED in historical OneHand162/Android profile	<code>lstm_features.py</code> , Android builder; FSL extractor keeps all pose landmarks
Velocity/delta features	INACTIVE	Android constant <code>USE_VELOCITY_DELTA_FEATURES=false</code> ; <code>buildWithDelta()</code> is dormant
225-feature onehand+delta contract	EXPERIMENTAL/CONFLICTING	<code>buildWithDelta()</code> pads to 225, while 225 normally means pose + two hands

Repository code supports the purpose of translation and scale normalization as reducing image-position and hand-size variation. It does not provide a formal empirical proof of the chosen 0.3 Z multiplier; treat its rationale as practical depth damping, not a validated optimum.

4. Temporal processing

Dynamic signs are not reliably identified by a single posture: motion direction, path, timing, and the relation between consecutive poses matter. VoxGest therefore classifies sequences, not MediaPipe labels.

FSL-105

- Sequence length: **20 frames**; feature shape: `(20,162)`.
- Window generation: all decoded frames are retained, then `range(0, len(vectors)-20+1, 5)` produces windows with **stride 5**.
- Quality gate: a window needs at least **13/20** frames with a detected right hand (65%).
- The source split is `train.csv_only`; FSL `test.csv` is not extracted by `76_extract_fs1105_features.py`.
- No velocity vector is concatenated to the FSL arrays. Temporal motion is supplied by the ordered coordinate sequence itself.

Android

- `LandmarkSequenceBuffer` holds **20** vectors and acts as a rolling/sliding buffer once full: each added frame removes the oldest.
- The controller invokes the recognizer whenever the full buffer is available and counts consecutive top-1 labels across sliding windows for its dynamic gate.
- A missing/invalid frame does not append. Eight lost-hand frames or 1.5 seconds without valid landmarks aborts the live collection; collection has an 8-10 second timeout depending on profile.
- Velocity delta calculation exists only behind the false `USE_VELOCITY_DELTA_FEATURES` flag. It is inactive and must not be assumed by a trained 162-feature model.

5. ML architecture and artifacts

FSL RD-TCN - EXPERIMENTAL / NOT TRAINED

`78_train_fsl_rdtcn.py` defines the exact intended model:

- Input: `(20,162)`; output: `C` softmax classes.
- Three causal residual blocks, each with two `Conv1D(filters=64, kernel=3)` layers, `LayerNormalization`, `ReLU`, and `SpatialDropout1D(0.2)`; dilation rates **1, 2, 4**. A 1x1 residual projection is used if channel widths differ.
- Global average pooling -> `Dense(64, ReLU)` -> `Dropout(0.3)` -> `Dense(C, softmax)`.
- Optimizer/loss: Adam `3e-4`, sparse categorical cross-entropy, accuracy metric.
- Training augmentation: temporal +/- one-frame shift, global scale 0.95-1.05, XY Gaussian noise (`sigma=0.005`), no Z noise, and re-zeroed nose triplet.
- Export: TFLite Optimize.DEFAULT; tries float16 conversion, then dynamic-range fallback. Android-ready criterion is validation accuracy ≥ 0.80 and model ≤ 500 KB.
- Parameter count: **UNKNOWN until an FSL model is built**; no `voxgest_rdtcn_fsl_v1.h5/.tflite` or FSL training report exists.

Existing saved model families

Family	Verified artifact contract	Architecture/status
OneHand162 phrase/calibrated	actual H5 and TFLite input <code>[1,30,162]</code> , output <code>[1,5]</code> , 273,837 parameters	Basic residual TCN, 96 filters, kernel 3, dilations 1/2/4/8, average+max pooling. EXPERIMENTAL; Android manifests incorrectly say 20 frames.
FullSign225 manual5 team v2	<code>[1,30,225]</code> -> <code>[1,5]</code>	Basic TCN experiment. Saved grouped validation 16.75%; not promotable.
FullSign225 RD-TCN	<code>[1,30,225]</code> -> <code>[1,5]</code> , 153,545 parameters	Experimental residual/dilated TCN, saved report 41.32% grouped validation.
Older demo/sprint LSTM and TCN	30-frame models and reports	IMPLEMENTED historical desktop experiments, not current FSL deployment.
Static alphabet	<code>[1,63]</code> -> <code>[1,29]</code> documented Android contract	Historical/disabled automatic Android output path.

No CNN/raw-video image model was verified in the current pipeline code. `Conv1D` is used as a temporal convolution, not as an image CNN. LSTM is present in legacy Python tooling and artifacts, but no FSL LSTM is defined.

6. FSL-105 data pipeline

```
64 import labels.csv (105 labels) -> generate_fsl_labels.py -> 64 confirmed one-handed label
> 75 classify_fsl105_handedness.py -> 64 confirmed one-handed label
> 76 extract_fsl105_features.py reads only train.csv rows for Short video
> MediaPipe Holistic -> normalized 162-feature frame
> 20-frame windows, stride 5, right-hand-presence as
> external_datasets/fsl_features/<LABEL>/fsl105_<ST>.npz
> 75 audit_fsl_dataset.py -> reports/fsl/audit_report.json
> 78 train_fsl_rdtcn.py (currently blocked by label contract)
> external H5, TFLite report, Android updates, Android tooling
```

Asset/stage	Current role
labels.csv	Dataset catalogue: 105 (id,label,category) rows. <code>generate_fsl_labels.py</code> writes an unsanitized 105-class JSON map, distinct from actual usable FSL training labels.
train.csv	1,704 source-video rows; the current FSL extractor uses this split only.
test.csv	426 source-video rows; present but not consumed by current extraction/training code.
class_labels_fsl.json	Generated mapping of all 105 label names to IDs. It is not consumed by <code>78_train_fsl_rdtcn.py</code> , which derives labels from <code>fsl_config.py</code> .
confirmed_onehanded_labels.json	64 classes accepted by the handedness workflow; controls <code>76_extract_fsl105_features.py</code> and <code>75_audit_fsl_dataset.py</code> .
.npz + .meta.json	Per-window float32 sequence plus label, source, <code>signer_id</code> , device, source video, window start, and hand-presence metadata.
75_audit_fsl_dataset.py	Requires <code>(20,162)</code> , finite values, valid metadata/device tag, and counts FSL-105 vs non-FSL device samples.
74_import_fsl_exports.py	Imports Android/phone JSON exports only if their stated shape matches <code>(20,162)</code> ; current report imported 0.

Train/test and leakage

The extractor protects the supplied FSL test split by not reading it (`train.csv_only`). That is a source-level train/test separation. However, extraction produces overlapping windows from a video (stride 5), so a later random window split would leak near-duplicate video content. The FSL trainer attempts a per-label signer holdout when at least two signer IDs exist; FSL windows all record `signer_id=fsl105_<video_stem>`, so this is effectively a video-stem holdout if multiple stems are present. It falls back to random 80/20 for a single signer, which is weaker and should be reported as leakage risk.

7. FSL-105 current status

Check	Actual status/evidence
Catalogue	105 labels in <code>labels.csv</code> ; 1,704 train and 426 test rows.
Handedness	64 one-handed, 41 two-handed, 0 ambiguous in <code>reports/fsl/handedness_report.json</code> .
Pipeline-recognized/extracted classes	64, defined by <code>confirmed_onehanded_labels.json</code> , not all 105.

Check	Actual status/evidence
Extraction	COMPLETE for configured source: 1,038 completed training videos, 16,146 saved windows.
Sequence shape	All generated FSL windows audit as (20,162); extraction summary reports zero shape mismatches, metadata errors, missing, unexpected, or orphan output files.
Class counts	Every 64 confirmed class has 180-366 FSL windows; all exceed audit minimum 100. No supplemental phone data is present.
Missing from onehand pipeline	41 two-handed labels are intentionally excluded. In addition, many desired FSL app labels (WHAT, YOUR, NAME, MY, NOTHING, WATER, etc.) have no FSL-105 onehand windows in this extraction.
Phone data	fsl_import_report.json: 0 imported. device_diversity_report.json flags 20 low-diversity labels.
Training readiness	NOT READY for the current FSL trainer. fsl_config.py lists 25 labels, of which only HELLO, YES, NO, and UNDERSTAND have extracted FSL-105 data and meet minima. The trainer requires at least five ready labels.
FSL model/TFLite/Android integration	NOT IMPLEMENTED; manifest status pending_training.

There is an older `extract_fs1105.py` that samples 30 frames of hand-only coordinates. It is a separate, incompatible legacy extractor and must not be confused with the current `76_extract_fs1105_features.py` contract.

8. Device generalization

The same physical sign recorded by PC, laptop, iPhone, or Android will not yield identical raw landmark numbers. Camera position, focal length and perspective, resolution, front/back-camera mirroring, crop, exposure, frame rate, pose estimator tracking state, partial occlusion, and MediaPipe's normalized image coordinate system all change the observations. Even a stable signer rarely repeats a trajectory identically frame-for-frame.

Implemented defenses are landmark representation rather than raw pixels; nose-relative translation; hand wrist-to-MCP scale normalization in the FSL/Android builder; Z damping; temporal windows; and selected-hand/pose masking. The FSL trainer also defines modest XY noise, temporal jitter, and global-scale augmentation, but these defenses are **not active until FSL training is actually run**. Device/signer diversity remains unresolved: FSL windows are tagged `FSL105_VIDEO0`, phone import is zero, and the audit identifies low-diversity labels.

9. Recognition logic

MediaPipe does **not** recognize a sign name. It returns detected landmarks. The sign label is produced only by a trained classifier operating on engineered temporal features.



For the Android five-label onehand path, `VoxGestTfliteRecognizer` ranks output probabilities, derives confidence/top-1 and margin (top1 - top2), and returns top-3. `DynamicWordAcceptanceGate` rejects blank/NOTHING/unsupported labels, wrong input shape, low hand presence, low confidence, low margin, insufficient consecutive matches, duplicates, and cooldown conflicts. Only accepted results are sent downstream; NOTHING is explicitly no-output. This logic exists, but cannot reach usable inference with the retained 30-frame model detected under the 20-frame profile.

No FSL class-probability thresholds, FSL decision gate, or FSL output token mapping were found.

10. Important engineering decisions

Decision	Supported rationale	Alternative considered/not preferred	Status
MediaPipe landmarks	Reduces raw-image dependence and provides pose/hand geometry to both Python and Android.	Raw video CNN not found in the current implementation.	IMPLEMENTED
162 features	Compact pose+selected-hand temporal representation.	FullSign225 preserves both hands but is a separate experimental profile.	IMPLEMENTED
Nose anchoring	Removes absolute image translation relative to a facial body reference.	Absolute image coordinates are not used in current FSL/Android feature builders.	IMPLEMENTED
Wrist-to-MCP scale	Reduces selected-hand scale variation.	No repository evidence of another scale rule for FSL/Android.	IMPLEMENTED
Z x 0.3	Explicitly configured in FSL/Android builders to reduce depth contribution.	No validation study found.	IMPLEMENTED, rationale partly UNVERIFIED
Temporal 20-frame FSL/Android buffer	Motion signs require evidence across time; matches Android buffer and new FSL pipeline.	Retained 30-frame models conflict with it.	IMPLEMENTED code, deployment CONFLICT

Decision	Supported rationale	Alternative considered/not preferred	Status
TCN/RD-TCN	Temporal Conv1D supports causal/dilated sequence processing.	Legacy LSTM remains available; FSL uses RD-TCN plan.	TCN IMPLEMENTED historically; FSL RD-TCN EXPERIMENTAL
Local TFLite inference	Android has local interpreter/loading and no backend requirement.	Cloud inference not implemented.	IMPLEMENTED infrastructure
FSL-105 one-hand subset	Avoids training a one-hand vector on two-handed signs.	FullSign225 exists for two-hand experiments but not FSL wiring.	IMPLEMENTED extraction policy
Velocity deltas	Code exists for a 225-wide output.	Flag is false; incompatible with current OneHand162 training contract.	INACTIVE

11. Current file map

FSL and active Android/FSL-adjacent files

File	Purpose	Input -> output	Connected component	Status
scripts_ml/fsl_config.py	FSL constants and 25 desired labels	config -> shapes/minima	FSL import/audit/trainer	EXPERIMENTAL; label conflict
scripts_ml/73_setup_fsl_dataset_dirs.py	Creates FSL data folder layout	config -> directories	FSL collection	IMPLEMENTED utility
scripts_ml/74_import_fsl_exports.py	Imports phone JSON samples	JSON -> (20,162) NPY/meta	FSL audit	IMPLEMENTED; 0 imports
scripts_ml/75_audit_fsl_dataset.py	Verifies FSL/phone data	arrays/meta -> audit JSON	FSL readiness	IMPLEMENTED
scripts_ml/76_extract_fsl105_features.py	Train-only FSL-105 extraction	video -> NPY/meta	FSL dataset	IMPLEMENTED
scripts_ml/77_record_pc_webcam_fsl.py	PC collection of FSL exports	webcam -> JSON exports	FSL importer	IMPLEMENTED collection tool
scripts_ml/78_train_fsl_rdtcn.py	Defines FSL RD-TCN train/export	FSL features -> H5/TFLite/report	FSL model	EXPERIMENTAL; blocked
scripts_ml/84_classify_handedness.py	Earlier handedness classifier	metadata/video -> classifications	FSL selection	EXPERIMENTAL
scripts_ml/93_classify_fsl105_handedness.py	MediaPipe-driven FSL-105 handedness audit	FSL videos -> reports/confirmed classes	FSL extractor	IMPLEMENTED audit
scripts_ml/generate_fsl_labels.py	Builds 105-ID label JSON	labels.csv -> class_labels_fsl.json	metadata only	IMPLEMENTED utility; not trainer input
scripts_ml/extract_fsl105.py	Older hand-only FSL extractor	video -> 30 sampled frames	obsolete FSL branch	LEGACY/INCOMPATIBLE
scripts_ml/check_20f_npy_shapes.py	Quick shape checker	a separate dataset path -> console	Android data check	EXPERIMENTAL; path mismatch
scripts_ml/70_import_android_onehand_exports.py	Imports Android calibration JSON	JSON -> (20,162) NPY/meta	Android-calibrated model	IMPLEMENTED

File	Purpose	Input -> output	Connected component	Status
scripts_ml/71_audit_android_onehand_exports.py	Audits imported Android data	NPY/meta -> report	Android-calibrated model	EXPERIMENTAL; 30-frame assumption
scripts_ml/72_train_android_calibrated_onehand162.py	Trains Android calibrated TCN	Android NPY -> H5/TFLite/manifest	Android runtime	EXPERIMENTAL; 30-frame assumption
model/runtime_manifest_fsl_rdtcn_v1.json	FSL target contract	config -> manifest	future Android deployment	pending_training
reports/fsl/extraction_summary.json	Extraction integrity snapshot	current outputs -> 16,146 result	FSL status	GENERATED, verified
reports/fsl/audit_report.json	Per-label FSL/phone readiness	arrays/meta -> audit	FSL status	GENERATED, verified
reports/fsl/handedness_report.json	64/41 class partition	FSL video analysis -> classes	FSL selection	GENERATED

Historical/experimental Python pipeline files

File(s)	Purpose	Input -> output	Connected component	Status
13_download_target_words.py , 35_discover_expansion_dataset.py	Acquire/discover word videos	video sources -> local data	desktop corpus	LEGACY
16_record_manual_words.py , 41_record_manual_fullsign225_words.py , 34_record_motion_letters.py , 26_record_phrase_intents.py	Webcam collection	webcam -> landmark sequences	legacy word/fullsign/phrase branches	LEGACY/EXPERIMENTAL
17_test_word_accuracy.py , 21_diagnostic.py , 22_diagnostic_lstm.py , 33_live_word_test_logger.py , 34_compare_live_logs.py , 30_eval_static_alphabet_live.py	Evaluation/diagnostics	models/logs -> measurements	desktop prototype	LEGACY
18_extract_lstm.py , 28_extract_phrase_videos.py	Video feature extraction	videos -> sequences	30-frame desktop paths	LEGACY
19_train_lstm.py	LSTM word classifier	30x162 data -> H5/TFLite/report	desktop words	LEGACY
24_train_tcn.py	Basic TCN trainer	30x162/225 data -> H5/TFLite/report	multiple historical profiles	LEGACY/EXPERIMENTAL
51_train_residual_dilated_tcn.py	RD-TCN experiment	fullsign data -> H5/TFLite/report	FullSign225	EXPERIMENTAL
27_train_phrase_tcn.py , 65_train_phrase_v1_tcn.py	Phrase temporal TCNs	phrase sequences -> artifacts	phrase branch	EXPERIMENTAL/frozen
35_train_motion_letter_tcn.py	J/Z motion TCN	30x162 -> artifacts	alphabet extension	EXPERIMENTAL
32_train_static_landmark_v4.py	Static alphabet model	hand landmarks -> artifacts	static alphabet	LEGACY
20_webcam_dual.py	Desktop combined live recognizer	webcam -> tokens	demo10 prototype	LEGACY
23_archive_manual_samples.py , 25_sprint_status.py , 30_audit_recognition_dataset.py , 31_record_static_calibration.py	Dataset maintenance/readiness	local data -> audit/recordings	old hardening flow	LEGACY
36_bootstrap_fullsign225_negatives.py , 37_cleanup_fullsign225_video_dataset.py , 38_clean_fullsign225_video_dataset.py , 39_audit_fullsign225_team16_extraction.py , 40_extract_fullsign225_team16_features.py	FullSign225 preparation/extraction	videos -> 30x225 data/reports	two-hand experiment	EXPERIMENTAL

File(s)	Purpose	Input -> output	Connected component	Status
42_audit_fullsign225_manual15_dataset.py , 42_audit_fullsign225_manual115_dataset.py , 42_audit_fullsign225_manual16_dataset.py , 48_audit_fullsign225_manual15_team_features.py , 50_audit_fullsign225_manual15_anas_seed_features.py , 53_audit_fullsign225_manual15_team_v2.py	FullSign audits	arrays/meta -> reports	FullSign225	EXPERIMENTAL
47_merge_fullsign225_team_recorded_features.py , 49_prepare_fullsign225_anas_seed_features.py , 52_merge_fullsign225_manual15_team_v2.py , 60_merge_fullsign225_phrase_v1.py , 62_merge_onehand162_phrase_v1.py , 64_process_latest_phrase_recordings.py	Merge/prepare recordings	contributor data -> datasets	fullsign/phrase	EXPERIMENTAL
61_audit_fullsign225_phrase_v1.py , 63_audit_onehand162_phrase_v1.py , 66_live_demo_runtime_plan.py	Phrase quality/runtime planning	features/logs -> audits/plans	phrase branch	EXPERIMENTAL
collect_fsl_alphabet_seq.py	FSL alphabet sequence collection	webcam -> arrays	auxiliary FSL work	UNVERIFIED/no current wiring
lstm_features.py	Shared legacy feature and sequence utilities	landmarks -> 162/225 vectors	desktop/fullsign code	LEGACY but influential
frame_quality_gate.py , gesture_segmenter.py	Quality/endpoint utilities	landmark frames -> state/gates	live/phrase tools	EXPERIMENTAL
word_config.py , motion_letter_config.py , phrase_config.py	Legacy profile configuration	config -> labels/thresholds	desktop models	LEGACY/EXPERIMENTAL
token_composer.py , phrase_builder.py , phrase_v1_dataset_tools.py	Token/data helpers	accepted tokens/features -> text/datasets	desktop/phrase	IMPLEMENTED helpers; not FSL runtime
export_tflite_fixed.py	Export verifier utility	H5 -> TFLite	historical model export	IMPLEMENTED utility

12. AI/ML conceptual map



MediaPipe does not assign `NAME`, `HELLO`, or an FSL word label. The trained temporal classifier maps a landmark sequence to class probabilities.

13. Current state matrix

Component	Status	Evidence	Next action
FSL one-handed classification	IMPLEMENTED	64 confirmed labels, 41 two-handed	Preserve partition; review only if a label changes scope
FSL train-only extraction	IMPLEMENTED	1,038 videos, 16,146 validated windows	Freeze/copy report baseline
FSL 20x162 quality/integrity	IMPLEMENTED	zero shape/meta/integrity errors	Independently re-run audit before training
FSL trainer label alignment	BLOCKED	four ready labels vs trainer minimum five	Reconcile config/confirmed labels deliberately
FSL RD-TCN training/TFLite	NOT IMPLEMENTED	manifest <code>pending_training</code> ; artifacts absent	Train only after alignment/split review

Component	Status	Evidence	Next action
FSL Android profile/output gate	NOT IMPLEMENTED	no FSL asset or label wiring in Android source	Add only after validated FSL artifact exists
Android 20-frame collection/features	IMPLEMENTED	buffer/builder/controller	Live-device parity test
Android onehand model deployment	BLOCKED	actual TFLite [1,30,162] , manifest [1,20,162] ; guard blocks inference	Train/export a matching model and verify TFLite tensor
FullSign225 models	EXPERIMENTAL	multiple saved models; weak validation for v2	Keep isolated; do not promote
Legacy LSTM/demo10/static paths	IMPLEMENTED historically	scripts/artifacts/reports	Do not confuse with FSL/20-frame runtime
Device diversity	IN PROGRESS	only FSL105 device; phone import zero	collect/import multi-device data

14. Next safe development order

1. Preserve the current FSL extraction report and verify the 64-label (20,162) dataset with `75_audit_fsl_dataset.py`.
2. Resolve the explicit FSL mismatch between `fsl_config.py` (25 target labels) and `confirmed_onehanded_labels.json` (64 extracted labels). Choose a documented training label set; do not silently rename `THANK YOU / THANKYOU` variants.
3. Verify that the future validation split is by source video/signer and that overlapping windows from one video cannot cross splits.
4. Audit class balance and source/device diversity; identify whether FSL only is sufficient for the intended first model and which phone captures are needed.
5. Train the FSL RD-TCN baseline only after steps 2-4; retain the generated confusion matrix, per-class accuracy, parameter count, TFLite size, and label order.
6. Validate the exported FSL TFLite input/output tensors against its manifest and labels.
7. Separately repair the existing Android onehand 20-vs-30 frame deployment mismatch; do not claim the Android model is live until an actual TFLite inspection passes.
8. Add an FSL Android profile, feature parity test, and FSL gate only after a validated FSL model exists.
9. Run controlled live tests across PC/laptop/iPhone/Android conditions; compare failure modes and update data/thresholds from evidence.

15. VOXGEST - AI CONTEXT HANDOFF

PROJECT STATE: VoxGest is a landmark-based temporal sign project with legacy 30-frame ASL/w ord/fullsign experiments, an Android 20-frame OneHand162 runtime, and a new FSL-105 extraction branch. Do not treat all artifacts as one compatible system.

CURRENT ARCHITECTURE: FSL target path is video -> MediaPipe Holistic pose/right hand -> nose-relative 162-vector -> (20,162) sliding windows (stride 5) -> future RD-TCN -> TFLite -> Android. Android code is camera -> MediaPipe -> OneHand162 builder -> 20-frame rolling buffer -> TFLite -> probability/margin/consistency gate -> accepted token.

FEATURE FORMAT: 162 float32 values/frame = 33 pose x 3 (99) + one hand x 21 x 3 (63). FSL/Android builder: nose-relative pose+hand, hand scaled by wrist-to-middle-MCP distance if >0.001, all Z values multiplied by 0.3; missing right hand becomes zeros in FSL extraction.

TEMPORAL FORMAT: FSL uses exact (20,162) windows, stride 5, >=13 right-hand frames. Android buffer is 20 frames and sliding. Velocity deltas are inactive.

DATA PIPELINE: FSL `labels.csv` (105) + `train.csv` (1,704) -> handedness partition -> 64 confirmed one-handed labels -> `76_extract_fsl105_features.py` -> `external_datasets/fsl_features` NPY/meta -> `75_audit_fsl_dataset.py` -> intended `78_train_fsl_rdtcn.py` -> intended FSL model/TFLite/Android. `test.csv` (426) is not used by current FSL extraction.

MODEL PIPELINE: Planned FSL RD-TCN: causal residual Conv1D blocks (64 filters, dilations 1/2/4), global average pooling, Dense 64, softmax; Adam/sparse CE; float16 or dynamic-range TFLite. No FSL model exists yet. Existing onehand artifacts are Basic TCNs built for (30,162).

FSL-105 STATUS: Extraction complete: 1,038 train videos, 16,146 (20,162) windows, 64 classes, zero reported shape/meta/integrity errors. Training is blocked because `fsl_config.py`'s 25 labels yield only four ready extracted labels (HELLO, YES, NO, UNDERSTAND) and trainer requires five. Phone import is zero; FSL test videos are unused.

IMPORTANT FILES: `76_extract_fsl105_features.py`, `75_audit_fsl_dataset.py`, `78_train_fsl_rdtcn.py`, `fsl_config.py`, `reports/fsl/extraction_summary.json`, `reports/fsl/audit_report.json`, `reports/fsl/confirmed_onehanded_labels.json`, `OneHand162FeatureBuilder.kt`, `LandmarkSequenceBuffer.kt`, `VoxGestTfliteRecognizer.kt`, `RecognitionProfile.kt`.

IMPLEMENTED COMPONENTS: FSL handedness/extraction/audit; Android camera/MediaPipe feature builder/buffer/gates; historical LSTM/TCN/fullsign tooling.

EXPERIMENTAL COMPONENTS: FSL RD-TCN, FSL phone data, FullSign225, phrase/motion-letter branches, Android calibrated onehand profile.

KNOWN PROBLEMS: FSL training label mismatch; no FSL model/deployment; Android asset manifests say 20 frames while actual bundled onehand TFLite tensors are 30 frames, and the runtime blocks this mismatch; device diversity is limited; legacy `lstm_features.py` normalization differs from FSL/Android builder.

DO-NOT-CHANGE CONSTRAINTS: Preserve existing data/model/runtime files unless separately authorized. Do not retrain, relabel, or merge pipelines silently. Treat status/shape conflicts as explicit engineering issues.

NEXT SAFE ACTION: Deliberately align and document the FSL training label set with the verified 64-class extraction, then audit source-video split integrity before any FSL training.